

Voyager AI — Business Document

1. The User

Primary: Independent travel advisors and small boutique agencies (1–10 people)

These advisors manage 20–100 active clients at a time. Each itinerary involves pre-paid tickets, confirmed hotel deposits, and restaurant reservations that cannot be moved without financial penalty. They charge \$150–500 in planning fees per trip and compete on two things: itinerary quality and response speed when something goes wrong mid-trip. A client texting at 9pm because their Day 3 museum is closed cannot wait until morning.

Secondary: Self-serve leisure travelers

Individuals planning honeymoons, family vacations, or international trips who want a structured, routable itinerary — not a prose list from ChatGPT. They plan 2–4 trips per year and will pay a small fee for something they can actually navigate.

2. The Problem

For advisors, building an itinerary today means hours across Google Maps, TripAdvisor, booking platforms, and spreadsheets. The deeper problem is disruptions. When a venue closes or a flight is delayed mid-trip, the advisor must mentally hold the entire schedule, identify which slots are locked (pre-paid, non-refundable), and rebuild a coherent day in minutes while the client is waiting on the phone. No existing tool handles this. ChatGPT will cheerfully suggest canceling a confirmed Colosseum ticket. Every current workflow — copy-paste into a doc, manual re-sequencing, searching for alternatives — is done by hand, under time pressure, for a client who is already frustrated. **For self-serve travelers**, AI-generated itineraries are unverified prose: closed venues, wrong opening hours, no routing logic, no way to adjust when plans change. They need something they can hand to a taxi driver and actually navigate.

Voyager AI produces verified itineraries from live Google Places data with real routing, respects pre-paid commitments as hard locks, and re-plans surgically in seconds.

3. The Economics

Pricing:

- Individual advisor: \$99/seat/month
- Agency (5+ seats): \$79/seat/month
- Self-serve traveler: \$12/itinerary

Cost to serve — one advisor-month (back of envelope):

A typical advisor runs 50 itineraries and 100 re-plans per month. Each planning run calls 6–7 agents across 20–30 LLM turns; each re-plan calls 1–2 agents across 5–8 turns.

Cost item	Basis	Monthly
LLM — Gemini 2.5 Flash (~3.75M tokens)	\$0.15/1M input, \$0.60/1M output	\$0.56
Google Places API (~400 calls)	\$0.017/call	\$6.80
Tavily booking search (~750 calls)	\$0.005/call	\$3.75
Cloud Run infrastructure	~\$5 flat for low-concurrency	\$5.00
Total cost to serve		~\$16

Revenue per seat: \$99. **Gross margin: ~84%.**

Self-serve: \$12/itinerary, ~\$3.05 cost to serve, ~75% margin.

Where it breaks:

Google Places is the largest cost driver at 43% of COGS. This is mitigated by the session-level candidate pool cache: Places results fetched during planning are stored in the session context and reused for re-plans, cutting re-plan Places costs by ~90%. LLM costs spike if re-plans are abused — a rate limit of 200 re-plans per month per seat caps downside. At a \$200 customer acquisition cost (advisor conferences, referral program), payback is month 3.

4. Why These Technical Choices

- **Multi-agent architecture** maps directly to advisor quality requirements. A single prompt producing lodging, activities, dining, and routing simultaneously produces lower-quality output and conflates concerns that have different failure modes. Separate specialist agents — one for hotels, one for attractions, one for restaurants — allow each to apply domain-specific ranking logic and fail independently. A bad restaurant recommendation is recoverable; a hallucinated hotel that the client cannot check into ends the advisor relationship. The architecture reflects this asymmetry.
- **Code-level locked-slot enforcement** is the core technical differentiator. Advisors manage non-refundable commitments: pre-paid Colosseum tickets, confirmed hotel deposits, reserved restaurant tables. If re-planning moves a locked slot, the advisor loses money and the client loses trust. Enforcing locks in Python — not just via LLM prompt instruction — provides a hard guarantee regardless of model behavior. This cannot be replicated by prompting a general-purpose model.
- **Candidate pool reuse** stores Places API results in session context so re-planning draws from the already-fetched pool instead of calling the API again. Re-plans are the high-frequency operation for advisors — 100 vs. 50 new plans per month. Reusing the pool cuts re-plan API cost from ~\$0.30 to ~\$0.03, a 10× reduction that is decisive at scale where re-plans are the dominant workload.
- **Google Places + Tavily hybrid** uses Places for structured venue data (hours, coordinates, ratings, booking flags) and Tavily to fill missing booking links — particularly restaurant reservation platforms (TheFork, OpenTable, Resy) that vary by country. Tavily's search ranking surfaces the locally dominant platform automatically, keeping the integration globally applicable without country-specific hardcoding.
- **Open-Meteo for weather** adds weather-aware scheduling at zero marginal cost. Scheduling an outdoor excursion on a forecasted rainy day is a preventable advisor error. For destination weddings and outdoor itineraries this is not a nice-to-have — it is a professional standard. Open-Meteo's free forecast API (16-day window) covers most planning horizons; historical climate averages handle trips beyond the forecast window.
- **Gemini 2.5 Flash via Vertex AI** balances reasoning quality with token cost at \$0.15/\$0.60 per million tokens input/output. The orchestrator requires multi-step conditional reasoning across 20–30 turns per plan; 2.5 Flash handles this reliably at a cost that keeps the per-plan LLM spend under \$0.03. Specialist agents use 2.5 Flash Lite for simpler, single-concern tasks (find restaurants, rank hotels), cutting specialist token cost by ~40% without quality loss on narrower tasks.